PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Sunday Ode Abi
Program/Batch	B084H-Networkwalks
Date	11 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

All activities documented in this report were conducted strictly for educational, research, and cybersecurity learning purposes. Testing was limited to systems and devices that I personally own or those for which I have obtained the necessary authorisation and permission.

The tools, techniques, commands, and findings presented in this report are intended solely to demonstrate my understanding of network reconnaissance, information gathering, and security assessment. They must not be used to gain unauthorised access to systems, disrupt services, compromise data, or engage in any unlawful activity.

I accept full responsibility for my actions and the proper use of the knowledge acquired during these exercises. Neither the instructor, the authors, nor any organisation referenced in this report shall be held responsible for the misuse of the information presented.

Unauthorised access to computer systems is a criminal offence in many jurisdictions, even where no damage or disruption occurs. Such actions may result in legal prosecution, financial penalties, loss of employment, and other serious consequences.

2. Introduction

This report documents the practical cybersecurity activities carried out during Week 2 of my ongoing internship programme at Networkwalks. The exercises focused on two important areas of network security: domain footprinting and network scanning.

The first exercise involved performing footprinting and reconnaissance on the networkwalks.com domain using various tools available in Kali Linux, including WHOIS, Nslookup, DNSRecon, cURL, WhatWeb, and WAFW00F. These tools were used to gather publicly available information about the domain, examine its DNS records, identify its hosting and web technologies, and determine whether a Web Application Firewall (WAF) was present.

The second exercise involved scanning my local network using Zenmap, the graphical interface for Nmap, on a Windows computer. A ping scan was performed against the 192.168.56.0/24 network to identify active hosts and observe their network relationships.

Together, these exercises demonstrate two important stages of network reconnaissance: gathering information about a domain from publicly available sources and identifying active devices within a local network.

All practical activities were carried out using Kali Linux for domain footprinting and Windows with Zenmap installed for local network scanning. Each exercise is documented with the command or scan configuration used, the observed results, supporting screenshots, and an explanation of the security significance of each finding.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, cURL, WAFW00F, and DNSRecon. Each tool was used to gather a different type of information about the target domain and its web infrastructure.

First, I used WHOIS to obtain publicly available domain registration information. The results identified GoDaddy.com, LLC as the domain registrar and showed that the registrant information was protected by a privacy service, Domains By Proxy, LLC. The results also revealed the domain's name servers, including NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM, and NS29.DOMAINCONTROL.COM.

Next, I used WhatWeb to identify the technologies running on the website. The results identified Apache as the web server, WordPress 7.1.4 as the content management system, and WP Download Manager 3.3.76. Other technologies identified included Bootstrap 7.1.4 and jQuery 3.7.1.

Using Nslookup, I resolved the domain name to its corresponding IP address. The result identified 192.232.216.135 as the IP address associated with networkwalks.com.

I then used cURL with the "-I" option to inspect the website's HTTP response headers. The result returned an HTTP/2 200 response, indicating that the server successfully responded to the request. The headers also exposed information about the server and revealed a WordPress REST API endpoint through "/wp-json/".

Next, I used WAFW00F to identify whether a Web Application Firewall was protecting the website. The result identified ModSecurity (SpiderLabs) as the WAF protecting the domain.

Finally, I used DNSRecon to enumerate the domain's DNS records. The results revealed information about the Start of Authority (SOA), name servers (NS), mail servers (MX), address records (A), and text records (TXT). The TXT records included SPF information and a Google site verification entry. The scan also reported that no SRV records were found and returned no answer for the DNSSEC query.

These activities helped me understand how different reconnaissance tools can be used to collect publicly available information about a domain, identify its underlying technologies, and examine aspects of its DNS and web infrastructure.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows "ipconfig" command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected Ping Scan to identify active hosts.

The scan was performed against the 192.168.56.0/24 subnet. The results identified three live hosts:

- 192.168.56.100

- 192.168.56.101

- 192.168.56.1

The scan also identified two MAC addresses:

- 08:00:27:9E:D1:21

- 08:00:27:DC:F0:92

A total of 256 IP addresses were scanned, with 3 hosts found active. The MAC addresses identified by the scan were associated with Oracle VirtualBox virtual network interfaces.

After completing the scan, I opened the Topology section in Zenmap to view the discovered hosts and their network relationships. The topology displayed the three discovered devices connected within the local network.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to `192.232.216.135`	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed `/wp-json/`	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● **Low**
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● **Medium**
6	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● **Medium**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. **Keep software updated**
   CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
   DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
   Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

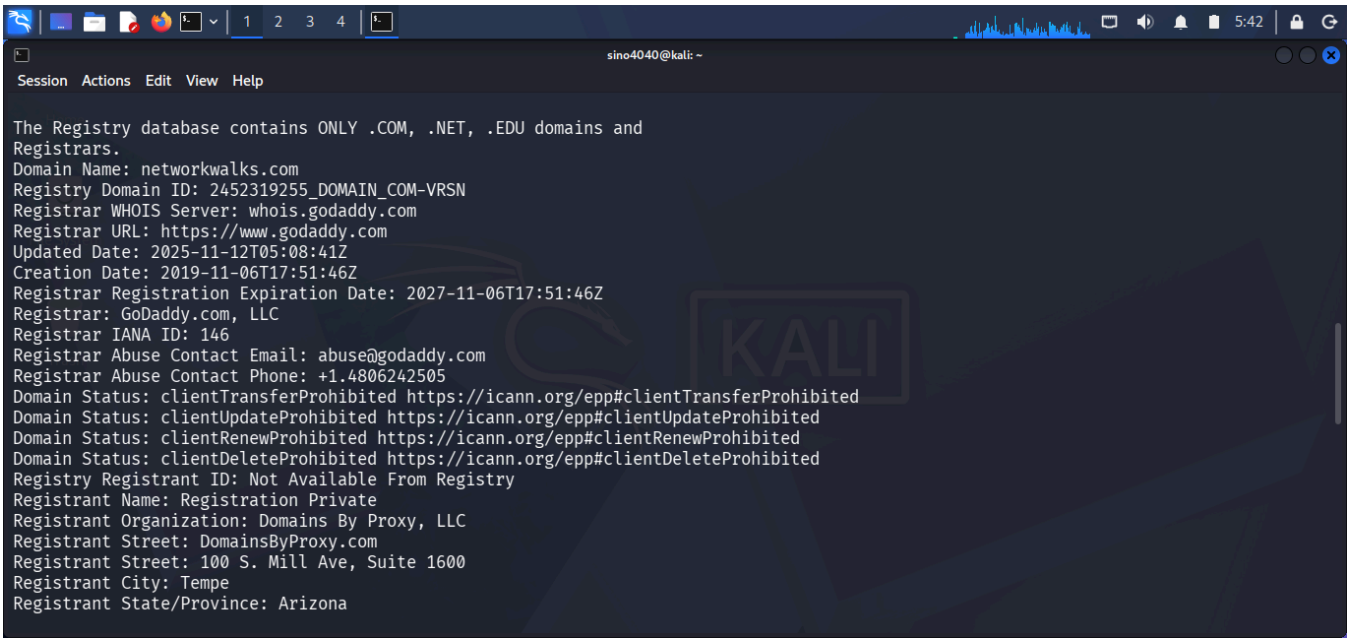
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

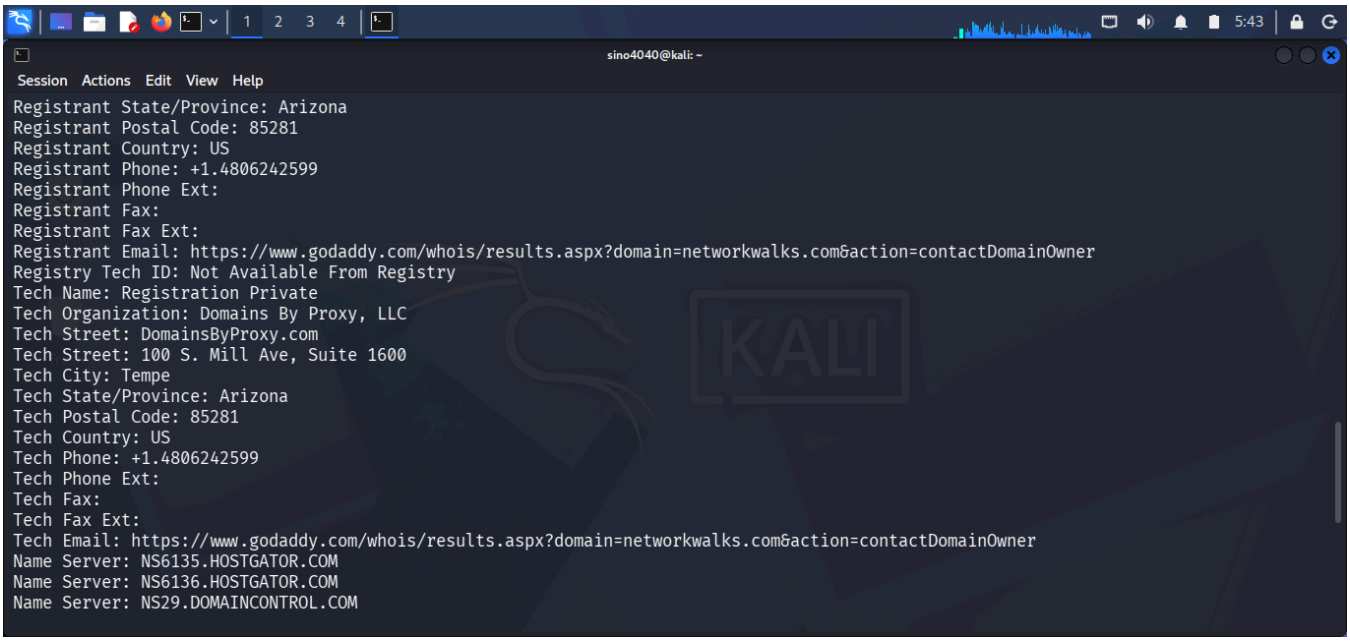
I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

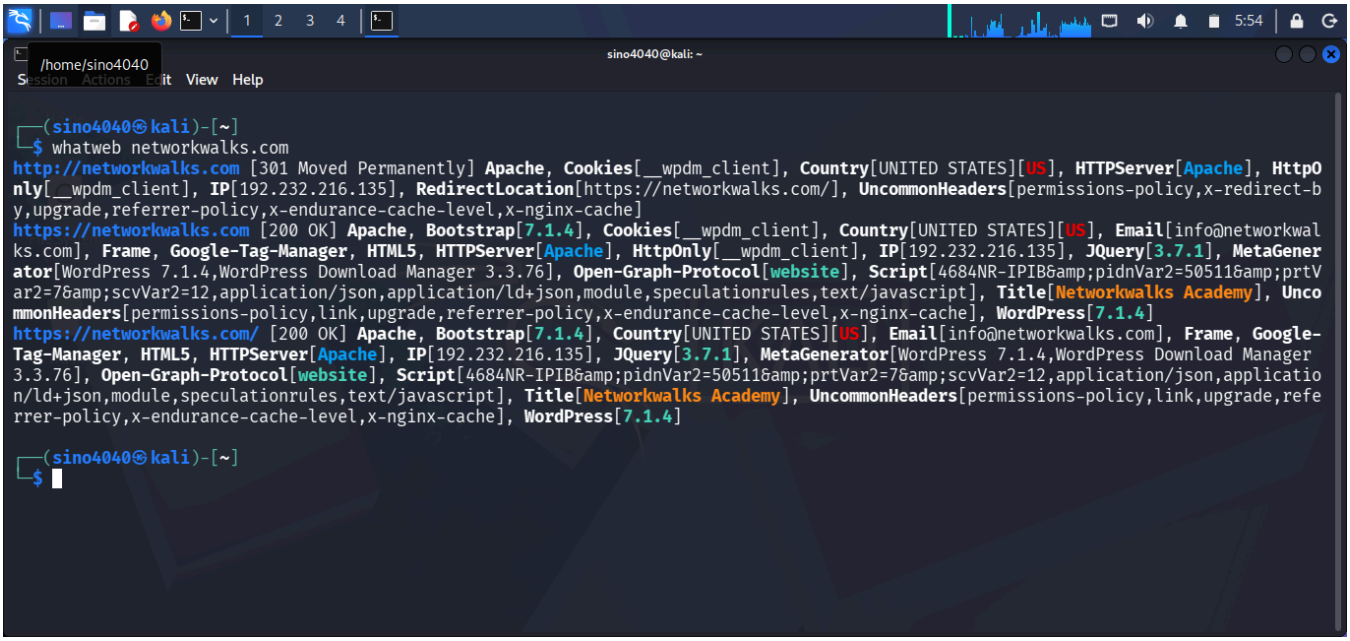
Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

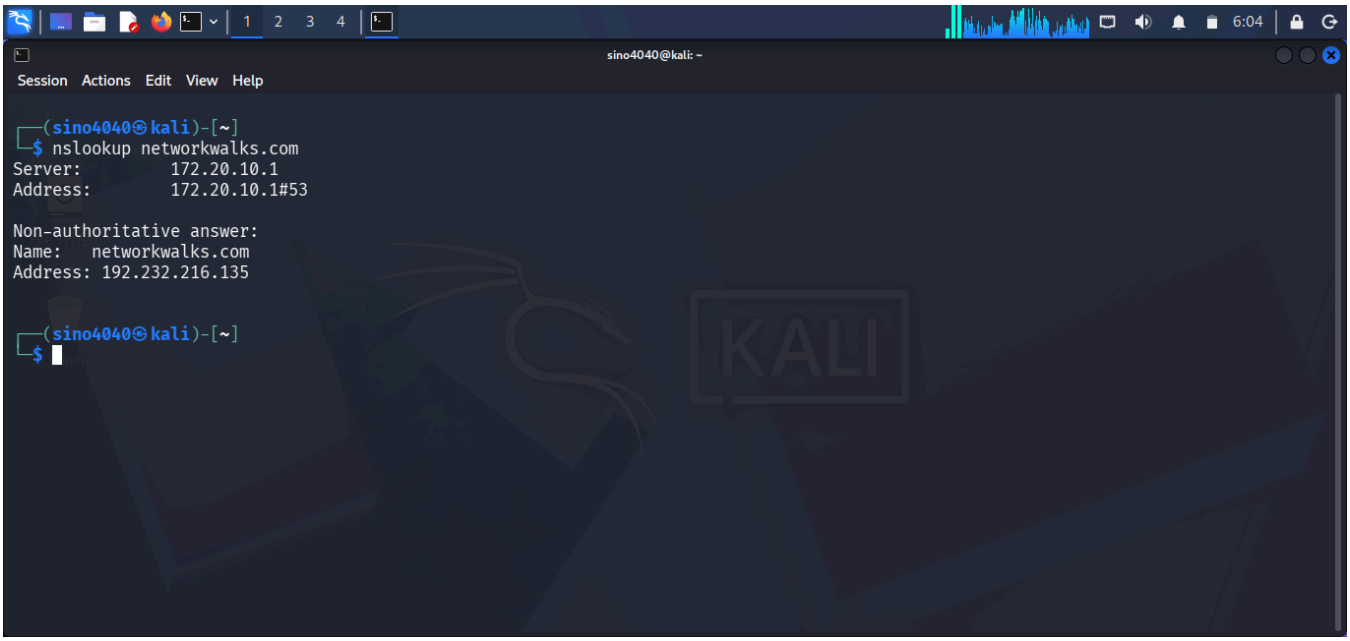
8. Evidences Collected



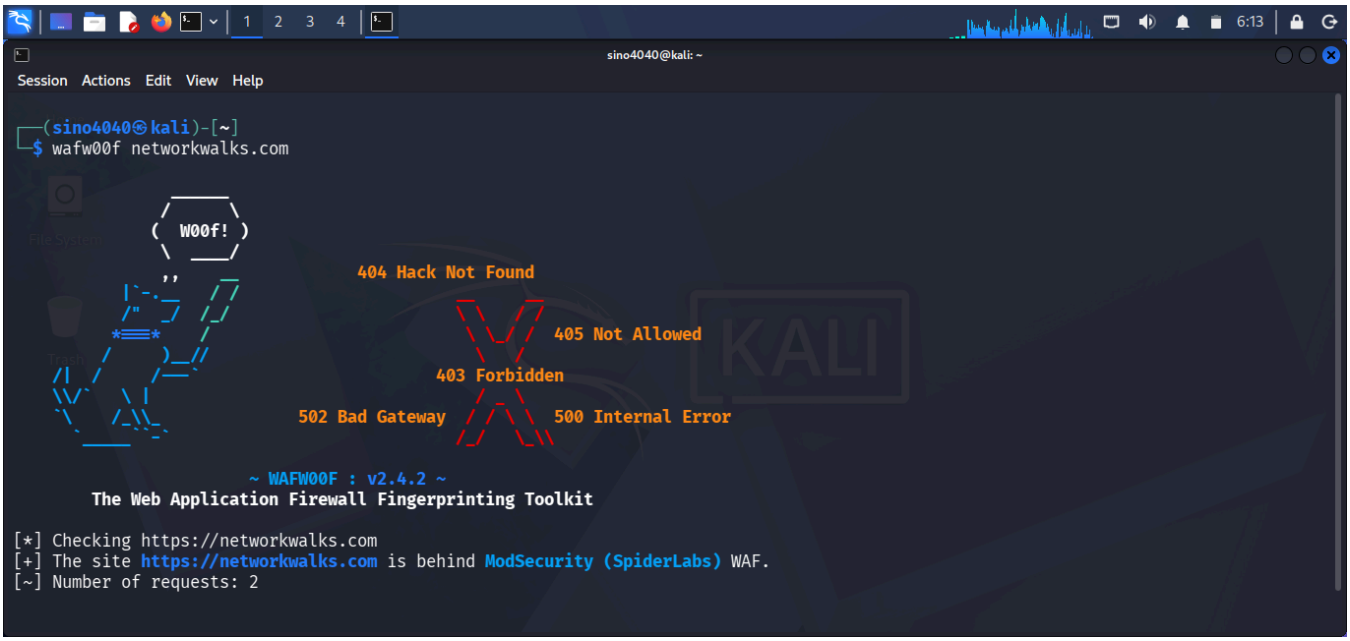


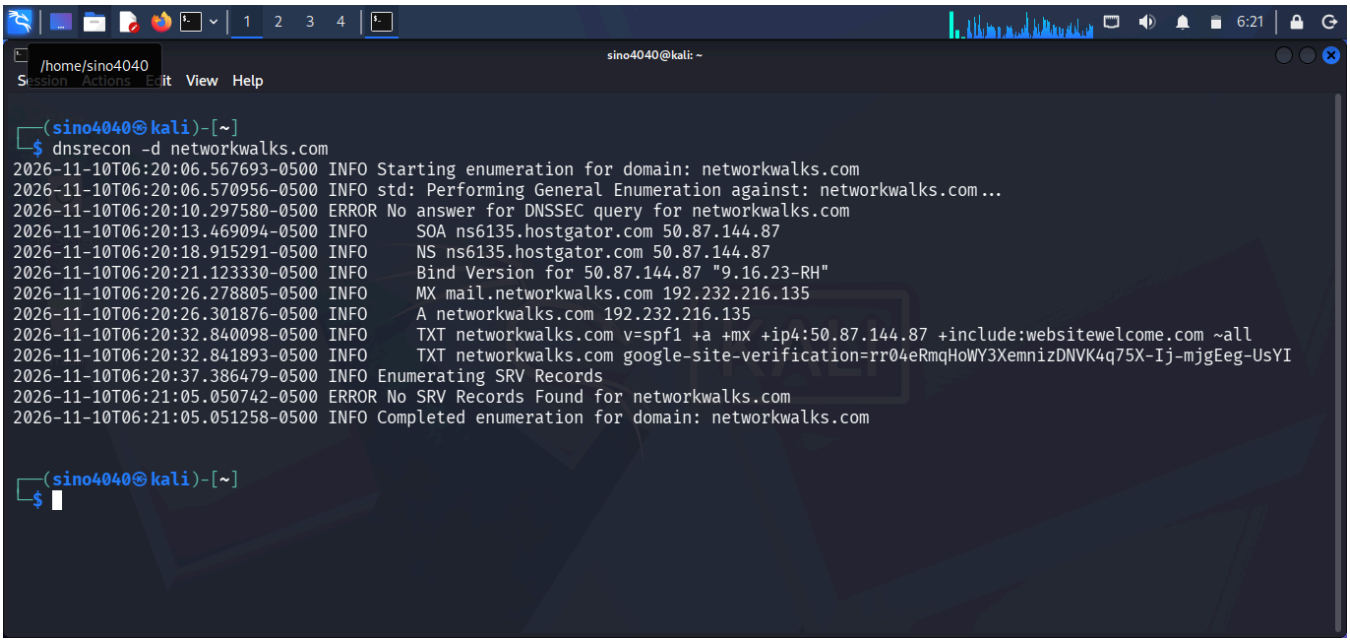

sino4040@kali: ~
Session Actions Edit View Help
Registrant State/Province: Arizona
Registrant Postal Code: 85281
Registrant Country: US
Registrant Phone: +1.4806242599
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Registry Tech ID: Not Available From Registry
Tech Name: Registration Private
Tech Organization: Domains By Proxy, LLC
Tech Street: DomainsByProxy.com
Tech Street: 100 S. Mill Ave, Suite 1600
Tech City: Tempe
Tech State/Province: Arizona
Tech Postal Code: 85281
Tech Country: US
Tech Phone: +1.4806242599
Tech Phone Ext:
Tech Fax:
Tech Fax Ext:
Tech Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
Name Server: NS29.DOMAINCONTROL.COM


/home/sino4040
sino4040@kali: ~
Session Actions Edit View Help
(sino4040㉿kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
(sino4040㉿kali)-[~]
$

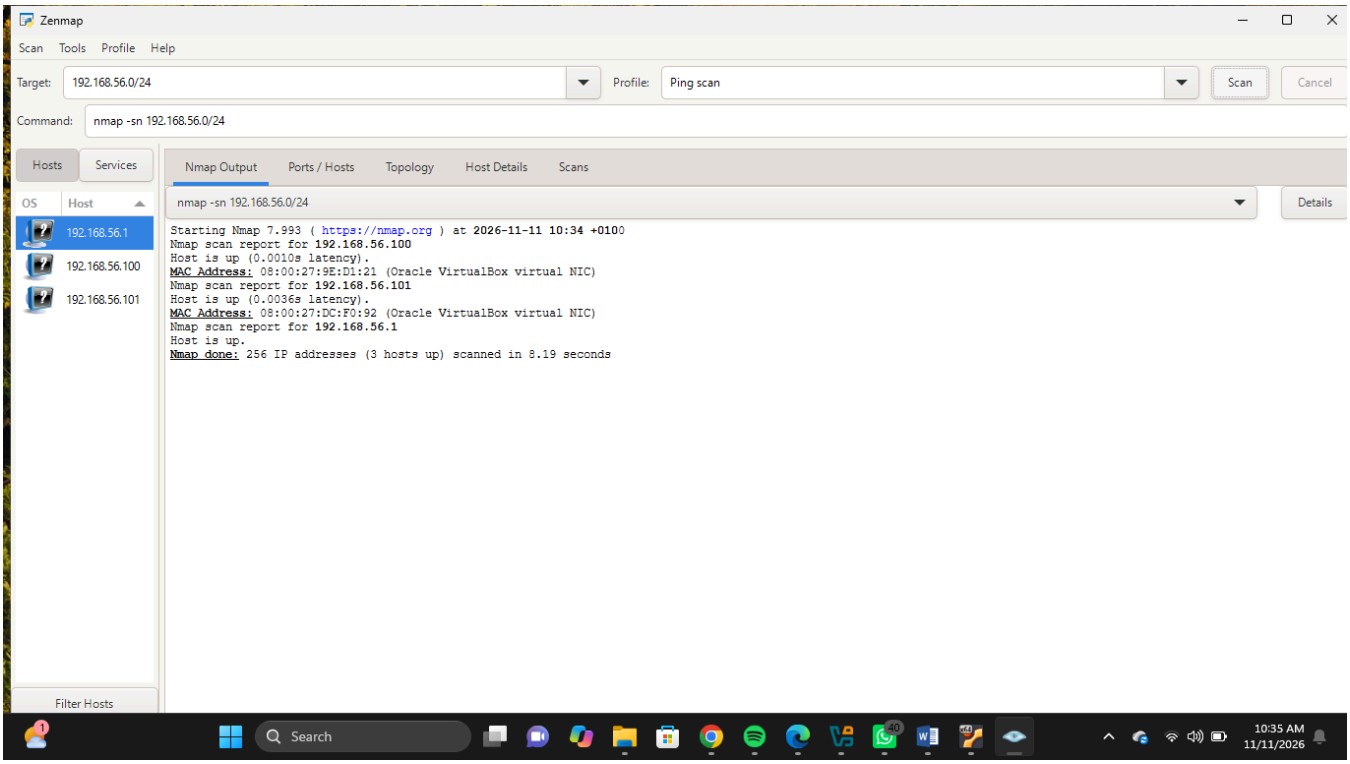

sino4040@kali: ~
Session Actions Edit View Help
(sino4040㉿kali)-[~]
$ nslookup networkwalks.com
Server: 172.20.10.1
Address: 172.20.10.1#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
(sino4040㉿kali)-[~]
$

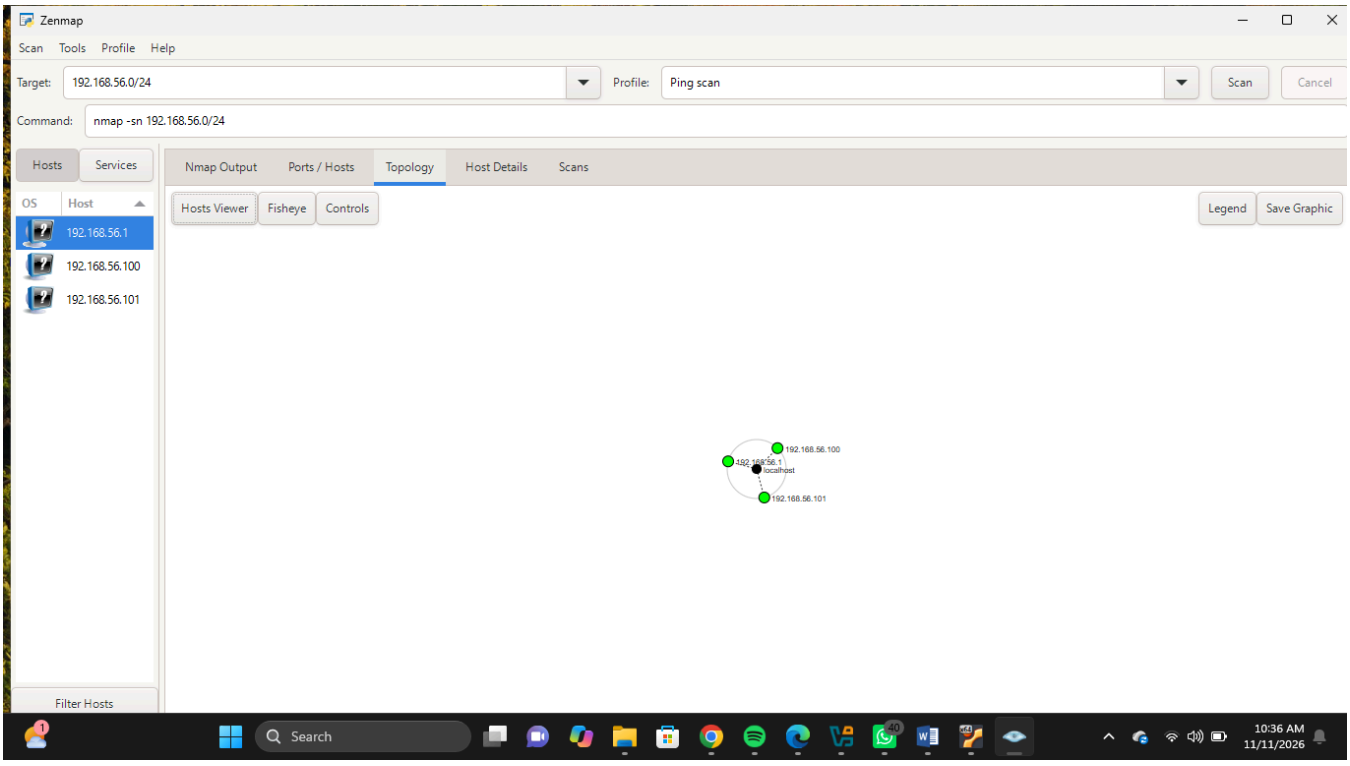

sino4040@kali: ~
Session Actions Edit View Help
(sino4040㉿kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=b50326391f6794cac4a32169f1515b94; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Tue, 10 Nov 2026 11:09:34 GMT
server: Apache
(sino4040㉿kali)-[~]
$ clear


sino4040@kali: ~
Session Actions Edit View Help
(sino4040㉿kali)-[~]
$ wafw00f networkwalks.com
W00f!
404 Hack Not Found
405 Not Allowed
403 Forbidden
502 Bad Gateway
500 Internal Error
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2


sino4040@kali: ~
/home/sino4040
Session Actions Edit View Help
(sino4040㉿kali)-[~]
$ dnsrecon -d networkwalks.com
2026-11-10T06:20:06.567693-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-10T06:20:06.570956-0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-10T06:20:10.297580-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-10T06:20:13.469094-0500 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-11-10T06:20:18.915291-0500 INFO NS ns6135.hostgator.com 50.87.144.87
2026-11-10T06:20:21.123330-0500 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-10T06:20:26.278805-0500 INFO MX mail.networkwalks.com 192.232.216.135
2026-11-10T06:20:26.301876-0500 INFO A networkwalks.com 192.232.216.135
2026-11-10T06:20:32.840098-0500 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-10T06:20:32.841893-0500 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-10T06:20:37.386479-0500 INFO Enumerating SRV Records
2026-11-10T06:21:05.050742-0500 ERROR No SRV Records Found for networkwalks.com
2026-11-10T06:21:05.051258-0500 INFO Completed enumeration for domain: networkwalks.com
(sino4040㉿kali)-[~]
$

My Nmap Scan Result





-End-

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📌 **Project Information**
Program Name: Cybersecurity Internship at Networkwalks | **Week:** 02 | **Repository:** GitHub